Balanced primes and primorial restrictions on consecutive runs

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Abstract

OEIS A006562 consists of the primes equal to the mean of their immediate prime neighbors. Its OEIS entry records conjectural asymptotics for the occurrence of these *balanced primes*. We establish an elementary, exact restriction on *runs* of balanced primes: when r consecutive primes are balanced, their $r+2$ neighboring primes form an arithmetic progression, and its step is divisible by every prime at most $r+2$, subject to a clearly stated small-prime exception. This supplies certified local information but no asymptotic density or infinitude theorem.

OEIS assertion and question

Write $p_1 = 2, p_2 = 3, \ldots$ for the increasing primes. The middle prime p_i is *balanced* precisely if

$$p_i - p_{i-1} = p_{i+1} - p_i \qquad (i \geq 2). \tag{1}$$

OEIS A006562 lists $5, 53, 157, 173, 211, 257, 263, \ldots$ and includes the conjecture

$$\lim_{n\to\infty} \frac{n(\log a(n))^2}{a(n)} = \frac{1}{2},$$

where $a(n)$ enumerates the balanced primes. We do *not* prove this conjecture. Instead we ask what restrictions are forced if many *successive primes* are simultaneously balanced.

Let $m^{\#} = \prod_{\substack{\ell \leq m \\ \ell \text{ prime}}} \ell$ denote the primorial.

Theorem 1 (Run–primorial obstruction)**.** *Fix integers $r \geq 1$ and $j \geq 1$. Suppose the r successive primes $p_{j+1}, \ldots, p_{j+r}$ all belong to A006562. Then there is a positive integer d such that*

$$p_{j+t} = p_j + td \quad (0 \leq t \leq r+1). \tag{2}$$

If, in addition, $p_j > r+2$, then

$$(r+2)^{\#} \mid d. \tag{3}$$

Proof. Apply (1) at indices $j+1, \ldots, j+r$. Each equality asserts that two adjacent gaps agree. Together they force the $r+1$ gaps from p_j through p_{j+r+1} to have the same positive value d, proving (2).

Let ℓ be any prime satisfying $\ell \leq r+2$, and assume for contradiction $\ell \nmid d$. The residues $p_j + td$ for $t = 0, 1, \ldots, \ell-1$ then form a complete system modulo ℓ. Thus one is divisible by ℓ. But all these numbers are among the primes in (2), and each is at least $p_j > r+2 \geq \ell$. None can equal ℓ; the divisible one would therefore be composite. This contradiction proves $\ell \mid d$ for every prime $\ell \leq r+2$, whence (3). □

Corollary 1 (Concrete gap restrictions)**.** *Apart from the initial balanced prime 5 with neighbors $(3, 5, 7)$, a balanced prime has both adjacent prime gaps divisible by 6. If three successive primes are balanced and their immediately preceding prime exceeds 5, their five-prime arithmetic progression has common difference divisible by 30.*

Proof. For one balanced center take $r = 1$ in the theorem: $3^{\#} = 6$ if the lower neighboring prime is greater than 3. The only possible lower prime at most 3 is 2 or 3; direct inspection of the initial prime triples $(2, 3, 5)$ and $(3, 5, 7)$ gives just the stated exception. For $r = 3$, take $5^{\#} = 30$. □

Corollary 2 (A fixed gap limits a run)**.** *Let q be the least prime not dividing d. Any run as in Theorem 1 with step d and preceding prime $p_j > r+2$ has length $r \leq q-3$.*

Proof. If $r \geq q-2$, then $q \leq r+2$ and Theorem 1 forces $q \mid d$, contradicting the choice of q. □

Checks and scope

The consecutive prime progression $(251, 257, 263, 269)$ has the two balanced centers $257, 263$ and common difference 6. A five-prime progression is

$$9843019, \quad 9843049, \quad 9843079, \quad 9843109, \quad 9843139,$$

whose gap is 30, in accord with Corollary 1; see the OEIS discussion of consecutive-prime arithmetic progressions [2]. These examples are illustrations, not ingredients of the proof.

Priority and limitations. The result is a direct specialization of the classical modular obstruction for prime arithmetic progressions. It is not presented as new, does not establish infinitely many balanced primes, and does not imply the conjectured A006562 asymptotic. It is a verified, sequence-specific structural lemma. This manuscript was AI-generated; symbolic checks and a proof review were performed, but no independent human peer review is claimed.

References

[1] OEIS Foundation, *A006562: Balanced primes*, https://oeis.org/A006562 (accessed 13 November 2026).

[2] OEIS Wiki, *Consecutive primes in arithmetic progression*, https://oeis.org/wiki/Consecutive_primes_in_arithmetic_progression (accessed 13 November 2026).